appropriate PCI message inputs for that optional system or equipment. Once a configurable indicator is enabled by the EMIC, it is learned and stored in cluster memory for the remainder of the cluster life.

The hard wired indicator inputs may be diagnosed using conventional diagnostic methods. However, the EMIC circuitry and PCI bus message controlled indicators are diagnosed using the EMIC self-diagnostic actuator test. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). Proper testing of the PCI data bus and the electronic message inputs to the EMIC that control an indicator requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information. Specific details of the operation for each indicator may be found elsewhere in this service information.

CLUSTER ILLUMINATION

The EMIC has several illumination lamps that are illuminated when the exterior lighting is turned on with the headlamp switch circuitry of the left multi-function switch. The illumination intensity of these lamps is adjusted when the interior lamps control ring on the left multi-function switch control stalk is rotated (downward to dim, upward to brighten). In response to that input, an analog/digital (A/D) converter in the EMIC converts the analog panel lamps dimmer resistor multiplexed input from the left multi-function switch into a digital dimming level message and a 12-volt Pulse-Width Modulated (PWM) output. The EMIC uses the PWM output to power the cluster illumination lamps and the VFD unit on the EMIC circuit board, then provides a synchronized PWM output on the hard wired fused panel lamp feed output circuit to control and synchronize the illumination intensity of other incandescent illumination lamps in the vehicle. The cluster illumination lamps are grounded at all times.

The EMIC also sends electronic dimming level messages over the PCI data bus to other electronic modules in the vehicle to control and synchronize the illumination intensity of their VFD units with that of the EMIC VFD unit. In addition, the interior lamps control ring on the left multi-function switch control stalk has a Parade Mode position to provide a parade mode. The EMIC monitors the request for this mode from the left multi-function switch, then sends an electronic dimming level message over the PCI data bus to illuminate all VFD units in the vehicle at full (daytime) intensity for easier visibility when driving in daylight with the exterior lighting turned on.

The hard wired left multi-function switch cluster illumination inputs and the EMIC fused panel lamp feed output may be diagnosed using conventional diagnostic methods. However, proper testing of the PWM control of the EMIC and the electronic dimming level messages sent by the EMIC over the PCI data bus requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

AUDIO SYSTEM CABIN EQUALIZATION

Each time the EMIC receives an electronic cabin equalization request message from the radio over the PCI data bus, it provides an electronic response to the radio containing the appropriate equalization curve information. Because there are numerous optional radios which are common to many platforms and available with various speaker architectures, each radio contains a Digital Signal Processing (DSP) microprocessor chip. This DSP chip uses the equalization curve information to optimize the radio's sound output characteristics for the unique cabin and speaker architecture found within the particular vehicle into which the radio has been installed.

Proper testing of the PCI data bus and the electronic data bus message inputs to and outputs from the EMIC that control audio system cabin equalization requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

AXLE LOCKER CONTROL

The EMIC contains a logic circuit and programming to perform the axle locker control functions for models equipped with the optional off-road package. The EMIC monitors hard wired inputs from the ignition switch on the fused ignition switch output (run-start) circuit and the key-in ignition switch on the key-in ignition switch sense circuit. The EMIC also monitors vehicle speed and transfer case shift position switch electronic message inputs received from the Powertrain Control Module (PCM) over the PCI data bus. The internal programming of the EMIC then determines whether to activate or deactivate the axle locker function by enabling or disabling the axle lock switch located in the instrument panel accessory switch bezel.

The EMIC programming controls the axle lock switch through two separate axle lock switch enable circuits, enable 1 and enable 2. In all cases, the EMIC will not activate either enable circuit if there are any transfer case shift position switch or vehicle speed sensor faults present. Whenever the ignition switch is in the On or Start positions, the key is in the ignition lock cylinder, the transfer case is in 4 X 4 Low range, and the vehicle speed is less than about 72 kilometers per hour (45 miles per hour) the first enable (enable 1) circuit is activated. The second enable (enable 2) circuit is activated only if the vehicle speed is less than about 16 kilometers per hour (10 miles per hour). When both enable circuits are activated, the axle lock switch becomes functional.

Once activated, the enable 1 circuit is automatically deactivated whenever the transfer case is moved out of the 4 X 4 Low range, or if the vehicle speed is greater than about 72 kilometers per hour